Shuyu Wang

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Research Interests

My research focuses on the dynamics and evolution of the Milky Way and the Magellanic Clouds, combining numerical simulations with observational data. I study past perturbations to the Milky Way disk using N-body+SPH simulations, focusing on the influence of the gas warp. I analyze the effect of interactions between the Magellanic Clouds on the morphology and kinematics of the Large Magellanic Cloud using N-body simulations. I compare simulation results with Gaia observational data.

Keywords: Milky Way dynamics, Magellanic Clouds, galactic dynamics, galactic archaeology

Education

Ph.D. Candidate in Astronomy Sep 2023 - Sep 2027 (expected)
Leiden Observatory, Leiden University, Leiden, The Netherlands
Supervisor: Dr. Anthony G.A. Brown
Thesis theme: Dynamics of the Milky Way and its Massive Neighbors

Master of Science in Physical Sciences Sep 2020 - Dec 2021
University of Chicago, Chicago, USA
Supervisor: Prof. Alexander P. Ji
Thesis: High-resolution Chemical Abundances of the Nyx Stream

Bachelor of Science in Physics Aug 2016 - Jun 2020
Southern University of Science and Technology, Shenzhen, China
Supervisor: Prof. Leonardo Modesto
Thesis: Analysis of Nonlocal Gravity in Terms of Auxiliary Fields

Undergraduate Exchange Student Aug 2019 - Dec 2019
University of California Berkeley, Berkeley, USA
Project paper: The Overview of Accretion Disk Models around Active Galactic Nuclei (AGN)

Publications

First Author Papers

Wang, S., Schölch, M., Romero-Gómez, M., Jiménez-Arranz, Ó., Brown, A.G.A., *Tidally-induced vertical asymmetries in the LMC bar and disk*, submitted to A&A

Wang, S., Brown, A.G.A., Debattista, V.P., Khachaturyants, T. 2026, *Phase Spirals Induced by the Gas Warp*, A&A, 710, A120

Wang, S., Necib, L., Ji, A.P., et al. 2023, *High-resolution chemical abundances of the Nyx stream*, ApJ,

955 129

Conference Proceedings

Hajnik, T., Padois, C., et al. (including **Wang, S.**) 2024, *Navigating the Stars: The MWGaia Doctoral Network's initiative to train the next generation of European astronomers in Gaia related science.*, EAS 2024, European Astronomical Society Annual Meeting, 1241

Research Projects

Tidally-induced vertical asymmetries in the LMC bar and disk 2026
Leiden University & University of Barcelona
Supervisors: Mercè Romero-Gómez & Anthony G.A. Brown

Phase spirals induced by the gas warp 2023 - 2025
Leiden University & University of Lancashire
Supervisors: Anthony G.A. Brown & Victor P. Debattista

Sensitivity of the CSST telescope for exoplanet microlensing event detection 2022
Tsinghua University
Supervisor: Wei Zhu

High-resolution chemical abundances of the Nyx stream 2021 - 2022
University of Chicago
Supervisors: Alexander P. Ji

Deep learning basics and strong lensing 2021
University of Chicago
Supervisor: Brian Nord

Accretion disk models around active galactic nuclei 2019
University of California Berkeley
Supervisor: Jessica Lu

Research Visits

University of Barcelona, Barcelona, Spain (hosted by Prof. Mercè Romero-Gómez) Jul 2026
Collaborated with Prof. Mercè Romero-Gómez and PhD candidate Marie Schölch on the analysis of vertical asymmetries in the LMC bar and disk using N-body simulations and mock Gaia DR3 catalogs

University of Lancashire, Lancashire, UK (hosted by Prof. Victor P. Debattista) Jan & Dec 2024
Collaborated on the detection and analysis of phase spirals using N-body+SPH simulations

Tsinghua University, Beijing, China (hosted by Prof. Wei Zhu) Feb - Jun 2022
Short-term research intern, collaborated on measuring the sensitivity of the CSST telescope for detecting exoplanet microlensing event

University of California Berkeley, Berkeley, USA (Summer school) Jul - Aug 2017

Grants and Awards

- MWGaiaDN: The Horizon Europe Marie Skłodowska-Curie Actions Doctoral Network 2023 - 2027
- Student Scholarship at the University of Chicago ($18 472) 2020 - 2021
- Outstanding Undergraduate Thesis 2020
- Second Class of the Merit Student Scholarship for Exceptional Performance (¥3 000) 2017 & 2018

Presentations

- **Phase spirals induced by the gas warp** (Poster with short presentation) Jun 2026
  European Astronomical Society Annual Meeting (EAS 2026), Lausanne, Switzerland
- **Phase spirals induced by the gas warp** (Poster) Jun 2026
  Galactic bars, bulges and disks workshop, Lipari, Italy
- **Phase spirals induced by the gas warp** (Contributed talk) May 2026
  Nederlandse Astronomen Conferentie (NAC 2026), Egmond aan Zee, The Netherlands
- **Warp-induced phase space spirals** (Flash talk) Aug 2025
  Winding, unwinding and rewinding the Gaia phase spiral (Lorentz workshop), Leiden, The Netherlands
- **Tilting Rate of the Milky Way Disc and Gaia Phase Spiral** (Short presentation) Mar 2025
  Netherlands Research School for Astronomy (NOVA) PhD school, Dwingeloo, The Netherlands
- **Tilting rate of the Milky Way disk** (Poster with short presentation) Jul 2024
  European Astronomical Society Annual Meeting (EAS 2024), Padova, Italy

Selected Workshops and Trainings

- **XVII Scientific Meeting of the Spanish Astronomical Society**, Tarragona, Spain Jul 2026
- **EDUCADO-MWGaiaDN training school on astro-AI and ML**, Ghent, Belgium Mar 2026
- **MWGaiaDN ESR School on transferable skills: managing, entrepreneurial and IPR**, Barcelona, Spain Jun 2025
- **MWGaiaDN: Galaxy modelling and galactic centre workshop**, Surrey, UK Dec 2024
  - Presented a group project *"Streams on Bar Resonant Orbit"* (Slides)
- **MWGaiaDN ESR school: introduction to astro-statistics and data visualisation**, Coimbra, Portugal Sep 2024
- **MWGaiaDN: The art of astrometry and computation**, Lund, Sweden Jun 2024
- **MWGaiaDN ESR induction school & diversity, public engagement and communication**, Leiden, The Netherlands Jan 2024

Technical Skills

- **Scientific programming:** Python, Shell/Bash, SQL, LaTeX, Git
- **High-performance and parallel computing:** multiprocessing, JAX, GNU Parallel
- **Galactic dynamics and modelling:** `AGAMA` and `galpy`; orbit integration, action/angle variables calculation, potential modeling, test particle simulation setup and analysis
- **Numerical simulations:** `pynbody` for N-body and N-body+SPH simulation snapshot processing and analysis, `GUDHI` for applying persistent homology to simulation data, performed Fourier decomposition on simulation data
- **Statistical inference:** Markov Chain Monte Carlo (MCMC) using `emcee`
- **Machine learning:** scikit-learn (dimensionality reduction using PCA, clustering using K-means), TensorFlow

Observing Experience

- Observing nights:
  - **Magellan (MIKE)**, 3 remote nights (PI: Alexander P. Ji) Oct 2021
- Stellar spectroscopy:
  - Used the radiative transfer code **MOOG** to obtain chemical abundances
  - Used **SMHR** to process Magellan (MIKE) and Keck (HIRES) spectra, fit equivalent widths, interpolate ATLAS model atmospheres, run MOOG, and perform spectral syntheses

Teaching Experience

- **Astronomical Observing Techniques** (B.Sc. course), Leiden University 2024 - 2027
  Teaching assistant, graded coursework, led weekly tutorial sessions (three semesters)
  Instructor: Prof. Nienke van der Marel

Academic Service and Outreach

- **Organizing committee member**, EDUCADO-MWGaiaDN training school on astro-AI and ML, Ghent, Belgium 2025 - 2026
- **Member**, RISE: Researchers in Science for Equality, Leiden University, The Netherlands 2023 - 2026
- **Intern**, African Program Office, UNESCO International Centre for Higher Education Innovation, Shenzhen, China 2019
- **Volunteer**, Shenzhen volunteer association, Shenzhen, China 2016 - 2020
- **Student union member**, Southern University of Science and Technology, Shenzhen, China 2016 - 2020

Language Skills

- English (fluent), Chinese (native)

References

- **Dr. Anthony G.A. Brown**
  brown@strw.leidenuniv.nl Leiden University
- **Prof. Victor P. Debattista**
  vpdebattista@gmail.com University of Lancashire
- **Prof. Mercè Romero-Gómez**
  mromero@fqa.ub.edu University of Barcelona